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10 QUESTIONS; 10 EXPERTS

Summary of the 2025 Professionals in Nutrition for Exercise and Sport "10 Questions/10 Experts" Session—Can Everyday Foods Replace Some Ergogenic Supplements and Commercially Available Sports Foods?

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This article provides a recap of the *10 Questions/10 Experts* session at the 2025 American College of Sports Medicine Annual Meeting. Each of the speakers considered the validity of common "myths" while providing evidence-based opinions to support or bust myths addressing the following questions: (a) Do people get enough dietary creatine from everyday foods? (b) Are nonsugar sweeteners an effective way to reduce energy intake in athletes? (c) Do beef jerky or chocolate milk provide enough leucine, making branched-chain amino acid supplementation unnecessary? (d) Can honey, applesauce, or other foods effectively replace carbohydrate gels for fueling exercise? (e) Is a homemade sports drink made from juice and a bit of salt just as effective as a commercially available sports drink? (f) Do vegetarians need to supplement carnosine and carnitine? (g) Is teff grain a rich source of iron? (h) Is coconut water a sufficient replacement for electrolyte drinks? (i) Can B vitamins influence the accuracy of urine color scoring to determine a low versus high urine concentration as a marker of hydration status? and (j) What is the impact of preexercise food-sourced calcium intake on acute bone metabolism in response to exercise? This article describes the content of each of the presentations including the most important outcomes and conclusions drawn by the presenters.

***Keywords**:* sports nutrition, sport foods, ergogenic aids, dietary supplements, food first

A highly anticipated feature of the American College of Sports Medicine Annual Meeting is the *10 Questions/10 Experts* session, hosted by Professionals in Nutrition for Exercise and Sport (PINES; https://pinesnutrition.org). Originally developed by Professor Ron Maughan, the session still utilizes a "speed dating" format, featuring 10 researchers or practitioners, who each tackle a current myth or controversy in performance nutrition under 3 min, before rendering a final verdict of "confirmed," "busted," or "plausible or equivocal." Plausible means believable, reasonable, or possible given the evidence, whereas equivocal means ambiguous or unclear. This engaging format encourages rapid communication of expert perspectives on complex topics.

This session has traditionally been led by Professor Louise Burke, and for the first time, Dr. Floris Wardenaar took up a moderator role this year, after Professor Emerita Melina Manore decided to step down. The 2025 edition continued its tradition of addressing timely and provocative issues in sports nutrition. This year's theme centered on a long-standing recommendation within the field—the "food-first" approach. However, the question remains, can everyday foods that are commonly consumed replace some ergogenic supplements and commercially available sports foods? Hence, the 2025 session turned its attention to the real-world feasibility of the food first, but maybe not always food only philosophy (Close et al., 2022), particularly in replacing ergogenic aids and commercially available sports products with whole or minimally processed foods. As this principle continues to be promoted across dietary guidelines and sports nutrition strategies, this session explored whether such substitutions could deliver equivalent performance benefits in practical settings. This article summarizes the key points presented by both emerging and established voices in the field.

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10 Nutrition-Based Questions Answered by 10 Experts

Do People Get Enough Dietary Creatine From Everyday Foods?

Eric Rawson, PhD, gave an answer to this statement by addressing the following proposition: "People consume adequate dietary creatine from everyday foods."

Creatine is a nonessential nutrient, and it is typically estimated that 1 g of creatine per day is produced endogenously and 1 g is ingested in the diet to satisfy needs. Dietary creatine is consumed primarily from meat and fish. Although there are differences in the creatine content of different cuts of meat and different types of fish, on average, there are about 4–5 g of creatine/kg of raw beef and fish. Relative to beef and fish, chicken is much lower in creatine (about 2 g of creatine per kilogram of raw chicken). Because high dietary creatine foods are animal flesh, vegan vegetarians consume almost no creatine, and other types of vegetarians (e.g., lacto-, ovo-, etc.) consume very small amounts of creatine from foods. Vegetarians have reduced blood and muscle creatine, and a brief vegetarian diet (~3 weeks) decreases muscle creatine. A 100 g (3.5 fl oz) serving of meat provides about 27 g of protein and about 0.5 g of creatine, so it seems as if the ingestion of 1 g of creatine per day would be easily achieved. However, one study of >89,000 people showed that dietary creatine intake decreased from 1999 to 2018, and two thirds of the population was ingesting <1 g of creatine per day (Korovljev et al., 2021). Recent data confirm that about two thirds of the population consume less than 1 g/day, which coincides with very low meat intake of ~20 g/day (Ostojic, 2021). This is not surprising, as the leading professional health and nutrition-related societies suggest that protein foods come from "mostly plant sources" (i.e., zero or very low creatine foods). Several investigations show that low dietary creatine intake is associated with poor health outcomes, including lower cognitive processing and worse memory (Ostojic et al., 2021). The available data suggests that dietary creatine intake is lower than the estimated need and associated with health consequences.

Was the Myth busted?

- Yes, this myth is busted.
- Explanation: The leading professional health and nutrition-related societies suggest that protein intake should come from "mostly plant sources" (i.e., zero or very low creatine foods) instead of meat. Further, population data show that dietary creatine intake is decreasing in children and adults, most people ingest <1 g/day (which is less than what is assumed), and low dietary creatine intake is associated with several indicators of poor health. An argument can be made that creatine should be considered a conditionally essential nutrient and that creatine supplementation might be prudent for optimal health.
- Additional reading: Ostojic and Forbes (2022).

Are Nonsugar Sweeteners an Effective Way to Reduce Energy Intake in Athletes?

Eimear Dolan, PhD, gave an answer to this statement by addressing the following proposition: "Nonsugar sweeteners are an effective means to reduce energy intake for athletes."

Nonsugar sweeteners (NSS), including synthetic compounds such as aspartame, sucralose, and saccharin, and naturally derived alternatives, such as stevia, provide a strong sweet taste with few to no calories. They are commonly used to replace free sugar in the diet without compromising taste. While most available NSS research focuses on health and obesity, they are receiving increased attention in athletic populations, where energy intake and physique management are often important concerns. Two relevant factors when evaluating the role of sweeteners in an athlete's diet are the potential for energy savings, along with other potential biological effects. The extent of energy savings ultimately depends on the amount of free sugar habitually consumed. For example, a teaspoon of sugar (~4 g) provides ~16 kcal. Therefore, substituting sugar in a single cup of coffee for a NSS is unlikely to make much difference to the athletes overall calorie intake. In contrast, replacing the ~38 g of sugar in a 335-ml (11.3 fl. oz) soft drink (which contains approximately 150 kcal), with a sugar-free artificially sweetened alternative (~0 kcal), may yield more meaningful energy savings across time. It is important to keep in mind, however, that food provides more than just calories. Diets high in added sugars may warrant broader evaluation of overall quality and nutrient adequacy, rather than focusing solely on potential calorie savings from substituting sugar for sweeteners. Furthermore, while habitually high dietary sugar may be a concern in general populations, athletes have specific nutrient and energy requirements, and sugar-containing foods and drinks can be an important source of carbohydrate (CHO) to support training and performance (Podlogar & Wallis, 2022). As such, athletes who are interested in using NSS to reduce their energy intake should take care to ensure this does not compromise CHO availability, and in turn, performance.

NSS are active biological compounds, with potential to impact various biological systems. Mechanistic studies have raised some concerns about the influence of certain NNS on gut microbiota and cardiometabolic health (Machek & Willoughby, 2019), although these studies generally use doses far in excess of those approved for human use. There are also some concerns that NNS might disrupt appetite regulation by decoupling sweetness from calorie intake, potentially promoting hunger and later compensatory overeating. Indeed, a recent study reported that acute sucralose ingestion increased hypothalamic blood flow—a purported MRI-based hunger marker—more than either sucrose or water in young adults (Chakravartti et al., 2025). How these findings translate to clinical or practical outcomes is, however, unclear. Extensive research has been conducted to evaluate the short and long-term influence of NNS use on energy intake, body mass, and health outcomes, but widely varying designs and often conflicting results render conclusive synthesis difficult. A recent World Health Organization commissioned systematic review and meta-analysis concluded that although some short-term controlled studies have reported positive NNS influence on energy intake, body mass, and body mass index, some prospective observational studies report the opposite and suggest a positive association between NNS use with obesity and health-related outcomes (Rios-Leyvraz & Montez, 2022; WHO, 2023). The observational nature of these studies means that causality cannot be determined, and it is not clear whether findings were influenced by factors such as altered gut microbiota or compensatory overeating, or were simply due to reverse causation. Nevertheless, the World Health Organization currently advises against using NSS for weight management or chronic disease prevention, due to uncertain long-term benefits and potential risks, although this stance has been criticized and calls for ongoing research have been made (Khan et al., 2023).

Was the Myth busted?

- No, the myth should be classified as equivocal.

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- Explanation: Overall, the evidence supporting the efficacy of NSS for managing energy intake and physique is largely inconsistent, with some findings suggesting potential benefits, while others raise concerns. Most of this evidence comes from studies in general or overweight/obese populations and, as such, may not necessarily be transferable to athletes, who have distinct energy and nutrient demands, metabolic responses, and behavioral contexts. Further research within a sports nutrition context is required before more definitive conclusions can be reached. Pending this research, while NSS may have utility in specific situations, such as athletes who struggle with calorie control but are unwilling or unable to reduce free sugar intake, current evidence indicates that they may not be an optimal primary strategy for energy or physique management for athletes.
- Additional reading: Machek and Willoughby (2019).

Do Beef Jerky or Chocolate Milk Provide Enough Leucine, Making Branched-Chain Amino Acid Supplementation Unnecessary?

Michael J. Ormsbee, PhD, gave an answer to this statement by addressing the following proposition: "A beef jerky stick or chocolate milk provides enough leucine, making BCAA supplementation unnecessary."

Branched-chain amino acids (BCAAs), comprising leucine, isoleucine, and valine, are three of the nine essential amino acids and play a pivotal role in muscle protein turnover (breakdown and synthesis; muscle protein synthesis [MPS]). Among the three BCAAs, leucine is regarded as the most potent activator of MPS due to its role in triggering the mechanistic target of rapamycin pathway, a key regulator of muscle anabolism. Research suggests that a leucine dose of approximately 2–3 g, commonly supplied by a single serving of BCAA supplements, is required to maximize MPS (Kaspy et al., 2023). However, recent data indicate that young women may respond to a much lower dose of 0.6–1.5 g of leucine (Apicella et al., 2025). BCAA supplementation is often used in exercising individuals aiming to enhance recovery, improve performance, and support hypertrophy. Some evidence suggests that BCAA supplementation may attenuate muscle soreness; however, most studies indicate that BCAAs alone have limited effects on overall performance and body composition, particularly when dietary protein intake is sufficient (Martinho et al., 2022). In such cases, the benefits of BCAA supplementation are significantly reduced. Nevertheless, individuals on calorie-restricted diets or with habitually low-protein intake may derive benefits from supplementation (Dudgeon et al., 2016). While supplementation is one method to increase BCAA intake, many whole foods, especially animal-based protein sources, are naturally rich in leucine. A standard beef jerky stick (28 g) contains approximately 0.5–0.7 g of leucine, while a 237 ml (8.0 fl oz) cup of chocolate milk contains 0.7–1.0 g of leucine, depending on the fat content of the milk. These foods provide enough leucine to reach the threshold required to maximize MPS for young women (0.6 g leucine), but not men (2–3 g leucine). Thus, for individuals seeking to optimize MPS, especially in scenarios of suboptimal protein intake, BCAA supplementation remains a more efficient strategy for some.

Was the Myth busted?

- No, this myth is plausible.
- Explanation: Beef jerky sticks and chocolate milk contain enough leucine for young women (0.6 g), but not enough for men (2–3 g), to acutely boost MPS. In addition, while beef jerky and chocolate milk may be eaten in abundance to meet the leucine threshold, the overall increased energy intake is a major consideration. For individuals in a caloric deficit or who follow a low-protein diet, BCAA supplements provide a more efficient way to reach this leucine threshold. However, it should be noted that high-quality protein (like whey) is greater than isolated BCAAs for building and repairing muscle.
- Suggested reading: Apicella et al. (2025).

Can Honey, Applesauce, or Other Foods Effectively Replace CHO Gels for Fueling Exercise?

Catalina Fernández-Campos, MS, gave an answer to this statement by addressing the following proposition: "Honey, applesauce, or other foods can effectively replace carbohydrate gels for fueling exercise."

The use of commercially available sports products pre or during exercise has increased exponentially over the last years. This increase is based on developments in food technology (i.e., modified starches) and applications in sport scenarios like road cycling, where consuming more than 120 g of CHOs per hour has been reported. Although it is appealing to use sports foods specifically tailored toward optimizing performance, a survey of 1,081 recreational, amateur, and professional endurance athletes showed that only 19% of the subjects relied solely in commercial sport products, while the majority, 67%, still preferred a mix of these products and everyday foods to reach their CHO needs for the energy demand during training (Reinhard & Galloway, 2022). According to Reinhard and Galloway, the main reasons for reporting the use of a combination of products are easier use and access, lower cost, a wider variety, and better gut feeling with everyday foods compared with sports foods. Some of the main foods used reported by the survey (i.e., fruit puree pouches, raisins, bananas, and potatoes) have been tested before for efficacy as described in previous literature. A review by Naderi et al. (2023) showed that natural CHO from everyday food sources can effectively improve performance in the same way as commercial sports products do. The use of everyday foods in place of sports foods during competition may require additional effort in planning and event logistics. Furthermore, there may be differences in gut response and comfort due to the concomitant intake of higher amounts of fiber or higher volumes of food to achieve the desired amount of CHO and a lack of information on CHO ratios within the product (Naderi et al., 2023). Given the increasing evidence that the gut can be trained to better absorb and tolerate CHO sources, there is potential for this to be applied to everyday foods to aid their use during exercise (Jeukendrup, 2017).

Was the Myth busted?

- No, the myth is confirmed.
- Explanation: Use of CHO-rich foods that are regularly consumed in the diet is just as effective in improving performance as commercially available sports foods, and they can effectively replace CHO gels, but the individual athlete's situation should be considered. Instead of choosing one strategy over the other, the combination of everyday foods and commercial products provides more opportunities for athletes who struggle with budget and the lack of variety in flavors or textures, or those who experience gastrointestinal disturbances and need

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different means to facilitate CHO intake. Overall, everyday foods may add to the nutritional toolbox that health professionals can use to tailor unique or specific athlete needs. More research is needed in different sports disciplines and intensities; but, for now, the use of everyday foods to replace sports foods may be a realistic strategy.

- Suggested reading: Naderi et al. (2023).

Is a Homemade Sports Drink Made From Juice and a Bit of Salt Just as Effective as a Commercially Available Sports Drink?

Stavros Kavouras, PhD, gave an answer to this statement by addressing the following proposition: "A homemade beverage can replace a commercially available sports drink."

A sports drink is a functional beverage specifically formulated for consumption during prolonged exercise to support rehydration, replenish electrolytes, and provide energy—primarily in the form of CHOs, to sustain high-intensity exercise. Hydration strategies during exercise often incorporate sports drinks to optimize training outcomes and enhance performance. Current recommendations for fluid composition during extended activity include CHO concentrations of 5%–10%, sodium concentrations of 20–30 mmol/L, and potassium concentrations of 2–5 mmol/L (Sawka et al., 2007). A key mechanism that facilitates rapid fluid absorption is the sodium-glucose cotransporter 1, which actively promotes the uptake of sodium and glucose in the small intestine (Hamilton, 2013). This cotransport system functions most efficiently when glucose and sodium are present in equimolar concentrations, typically around 13.5 g of glucose and 74 mmol (1.7 g) of sodium per liter of fluid. This formulation enhances water absorption and has informed the design of many commercial sports drinks. Over the years, athletes have developed homemade formulations based on these principles, aligning with the American College of Sports Medicine guidelines (Sawka et al., 2007).

One study by Shirreffs et al. (2007) compared the rehydration capacity of a commercial sports drink to a carbonated water and apple juice mixture (Apfelschorle), a beverage popular in Germany and high in potassium. The findings indicated that while Apfelschorle performed similarly to bottled water in terms of rehydration, it was less effective than the commercial sports drink. Beyond the basic composition guidelines for hydration beverages, several additional formulation factors should be considered. These include the type of CHO used (e.g., mono-, di-, or polysaccharides), gastrointestinal tolerance, flavor—which may influence voluntary fluid intake—and the inclusion of novel bioactive ingredients such as amino acids, which may affect fluid absorption, or caffeine, which may offer additional ergogenic benefits (Funnell et al., 2024). These components may enhance palatability, safety, and performance, particularly in individual or elite performance settings.

It is also important to consider that homemade CHO-containing drinks may pose a risk of microbial growth if not prepared and stored properly. Ultimately, the effectiveness of any hydration strategy is influenced by the individual's hydration status, physiological variability, gastrointestinal tolerance, and the nature and duration of the exercise. Therefore, beverage composition should be tailored to specific physiological and contextual needs. The collective evidence supports the use of physiologically informed hydration strategies that optimize solute composition to enhance fluid absorption, maintain fluid-electrolyte balance, and support both health and exercise performance.

Was the Myth busted?

- No, the myth is plausible.
- Explanation: Although a substantial body of research supports the efficacy of commercial sports drinks in promoting rehydration and improving exercise performance, limited data exist on the effectiveness of homemade alternatives. Nevertheless, many athletes and practitioners believe that, when appropriately balanced, simple homemade beverages—such as diluted fruit juice with added sodium—may offer hydration and performance benefits comparable to commercial products. A typical recipe might include 1 L of water, 60 g of sugar, honey, or other CHO sources, one-fourth teaspoon of table salt, and a splash of fruit juice for flavor.
- Suggested reading: Funnell et al. (2024).

Do Vegetarians Need to Supplement Carnosine and Carnitine?

Bryan Saunders, PhD, gave an answer to this statement by addressing the following proposition: "Vegetarians cannot get enough carnosine and carnitine from their diet to function normally."

Carnosine, a dipeptide synthesized from L-histidine and its rate-limiting precursor beta-alanine, is found predominantly in human skeletal muscle and functions primarily as an intracellular pH buffer (Saunders et al., 2017). Carnitine plays a critical role in cellular energy metabolism by facilitating the transport of long-chain fatty acids across the mitochondrial membrane for β-oxidation, an essential step in ATP production during prolonged exercise. Vegetarians, who do not consume meat, poultry, or fish—foods rich in beta-alanine (via histidine-containing dipeptides such as carnosine, anserine, and balenine also known as ophidine) and carnitine—may exhibit altered muscle levels of these compounds. Indeed, vegetarians have been shown to have lower skeletal muscle carnosine content compared with omnivores (Everaert et al., 2011). However, considerable interindividual variability exists, and some omnivores display carnosine levels comparable to or even lower than those of vegetarians (Saunders et al., 2017) though, importantly, no clinical carnosine deficiency has been reported in vegetarians. Interestingly, beta-alanine supplementation consistently increases muscle carnosine content and improves high-intensity exercise performance, with the magnitude of increase in carnosine appearing independent of baseline carnosine levels (Saunders et al., 2017). The impact of vegetarianism on carnitine status appears less clear. One study reported lower plasma and muscle carnitine levels and reduced carnitine uptake into muscle in vegetarians compared with omnivores (Stephens et al., 2011). However, another study found no baseline differences in muscle carnitine content, though 12 weeks of carnitine supplementation modestly increased muscle levels only in vegetarians (Novakova et al., 2016). Importantly, none of these studies demonstrated that a vegetarian diet leads to a clinically or functionally significant carnitine deficiency, suggesting that such diets do not inherently compromise carnitine status in healthy individuals.

Was the Myth busted?

- Yes, the myth is busted.
- Explanation: Vegetarians do not need to supplement beta-alanine or carnitine because they generally appear to achieve adequate muscle carnosine and carnitine without consuming meat-based sources. While levels might be a bit lower in vegetarians, there is a lot of individual variation, and it does not seem to make a meaningful difference to health or function. However, athletes—vegan, vegetarian, or omnivorous—seeking

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a competitive performance advantage may benefit from targeted supplementation with beta-alanine or carnitine.

- Suggested reading: Everaert et al. (2011).

Is Teff Grain a Rich Source of Iron?

Floris Wardenaar, PhD, gave an answer to this statement by addressing the following proposition: "Athletes should consume teff grain (Eragrostis tef) to obtain more dietary iron."

Teff, coming from Ethiopia and included in their national dish "injera," is claimed to be rich in iron (Fe). Furthermore, it has been branded a "superfood" that contributes to Ethiopian running success. In the United States, the Food and Drug Administration requires that a food must contain at least 20% of the daily value of Fe per serving, equaling to 3.6 mg based on 18 mg daily value of Fe, to support claims such as: "High in, an excellent source of, or rich in iron" (NIH Office of Dietary Supplements, 2025). Fe values ranging from 7 to 301 mg/100 g, and 2 to 94 mg/serving of teff have been reported (Dame, 2020; Guja & Baye, 2018; Seyoum et al., 2024). Traditional threshing of teff, in which cattle walk on the grain to separate the grain from the plant, results in substantial extrinsic soil Fe contamination (Guja & Baye, 2018). Comparing the Fe content of traditional threshed or laboratory threshed teff showed that in the absence of this contamination, the Fe content of the same grain sample was four times lower (Guja & Baye, 2018). Finally, considering the limited Fe bioavailability of extrinsic soil Fe, which may range from 2% in humans (Hallberg & Bjorn-Rasmussen, 1981), up to ~20% in rats (Guja & Baye, 2018), extrinsic Fe may lead to small elevations of bioavailable Fe. Guja and Baye (2018) reported Fe levels for a serving of teff (31 g or one-fourth cup) of 9.2 mg for traditionally threshed teff and 2.1 mg for laboratory threshed teff. When correcting for the limited bioavailability of extrinsic Fe (applying a maximum bioavailability of 20%), the traditionally threshed teff would contain ~3.5 mg of Fe per serving, which is slightly below the Food and Drug Administration definition for a Fe source. Applying the same corrective calculations for extrinsic Fe to literature reports of the Fe content of teff, four analyses of different teff varieties would result in Fe levels below this cut-off, while another five teff analyses reported levels that would meet the definition of a source rich in Fe (Dame, 2020; Guja & Baye, 2018; Seyoum et al., 2024).

Was the Myth busted?

- No, the myth is plausible.
- Explanation: The Fe content of teff grain depends on the Fe content of the soil in which it is grown and whether it has been threshed by traditional methods that allow contamination with the soil Fe to occur. Currently, most of the global teff supply is grown and harvested in Ethiopia, and if grown in soil high enough in Fe, it is possible that it may be sufficiently contaminated to be considered a rich source of Fe.
- Suggested reading: Guja and Baye (2018).

Is Coconut Water a Sufficient Replacement for Electrolyte Drinks?

Kinta Schott, MS, gave an answer to this statement by addressing the following proposition: "Athletes can replace an electrolyte drink with coconut water."

During prolonged aerobic exercise, athletes may lose as much as 1 L of body fluids per hour (Ismail et al., 2002). In addition to the fluids lost, sodium and potassium may also be lost, which may lead to electrolyte imbalances and subsequent health and performance detriments. It is recommended that athletes utilize (re)hydration strategies to maintain or replenish body fluids and electrolytes lost during and after exercise. Guidelines for fluid intake before, during, and after exercise are well described. Postexercise, 1.2–1.5 times the amount of fluid lost should be consumed (Ismail et al., 2002; Kalman et al., 2012; McDermott et al., 2017). Regular sport drinks often contain sodium, as well as potassium, and additional CHOs to help replenish electrolytes, body fluids, and glycogen lost during exercise (Ismail et al., 2002). While many athletes rely on these popular sport beverages, some search for more "natural" or palatable alternatives like coconut water (Ismail et al., 2002; Kalman et al., 2012). Fresh coconut water contains potassium, sodium, and chloride, and it has been used in clinical rehydration of patients experiencing severe dehydration due to diarrhea (Adams & Bratt, 1992). When compared with a commercially available electrolyte sports beverage, results are varied, and it appears coconut water may have similar rehydration abilities but some state it may be closer to plain water or may require additional sodium to provide benefits similar to sports beverages (Brier et al., 2025; Ismail et al., 2002; Ismail et al., 2007). Although fresh coconut water contained significantly higher levels of potassium (51 mmol/L) when compared with a sports beverage (3.5 mmol/L), coconut water's sodium levels were much lower than sport beverages, resulting in values of 9 versus 20 mmol/L, respectively (Ismail et al., 2007). Regarding rehydration, coconut water, especially when enriched with sodium, appears to perform not significantly different from commercial sports drinks across multiple hydration metrics, such as plasma volume, body mass regain, osmolality, and urine output (Kalman et al., 2012), although others did not show any differences for hydration markers between fresh coconut water and plain water (Brier et al., 2025). No meaningful differences in treadmill exhaustion or similar performance tests have been reported between coconut water and other drinks (Kalman et al., 2012). Tolerance and palatability seem to vary between studies with some reporting slightly less stomach upset in fresh coconut water, whereas others report mild stomach upset while using coconut water. Additionally, taste acceptance may vary; accessibility and cost may also play into the decision to use coconut water (Brier et al., 2025; Ismail et al., 2002; Ismail et al., 2007; Kalman et al., 2012).

Was the Myth busted?

- Yes, this myth is busted.
- Explanation: For athletes searching for a "natural" electrolyte beverage, coconut water with additional supplementation of sodium may be a viable option; however, without extra sodium, coconut water is more likely to mirror the hydration properties of plain water. Depending on personal preference, coconut water could help individuals to consume more fluid as taste may increase fluid intake compared with plain water, for example, after losing a large amount of fluid after prolonged exercise in the heat.
- Additional reading: Chaubey et al. (2017).

Can B Vitamins and Food Colorants Influence the Accuracy of Urine Color Scoring to Determine a Low Versus High Urine Concentration as a Marker of Hydration Status?

Yasuki Sekiguchi, PhD, gave an answer to this statement by addressing the following proposition: "B-vitamins and food colorants (e.g., beetroot) influence the accuracy of urine color scoring to determine a low vs. high urine concentration."

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Hydration is an important factor for health and performance, and the hydration assessment is essential to optimize body fluid balance (Armstrong, 2007). Urine color (Uc) is one of the most common methods to monitor hydration status as renal urine excretion is closely related to body fluid balance (Armstrong et al., 1994; Sekiguchi et al., 2023). It has been widely suggested that vitamin supplementation could make Uc more yellow and appear more concentrated. However, investigating the impact of B vitamins on Uc, the difference between conditions was too small to change the interpretation of hydration status (Kenefick et al., 2015). While there is limited data, Kenefick et al. (2015) examined the effect of B2, B12, and B2 + B12 supplementation on Uc in a euhydrated state. This study found that a large overnight vitamin supplementation did not change the hydration assessment for euhydration by using Uc (Kenefick et al., 2015). Although there were no statistical and practical differences, it is interesting to note that Uc was scored on average one scale darker (using an eight color Uc chart) when vitamins were supplemented compared with a control without supplementation (Kenefick et al., 2015). Another study indicated a one-unit difference in average Uc scores between riboflavin supplementation versus controls (McKenzie et al., 2017). Despite the potential of vitamin supplements resulting in a darker Uc, it does not have to affect Uc scoring accuracy. A stratified analysis to assess the accuracy of Uc demonstrated that an athletic population reporting the use of dietary supplements did not report a different accuracy than the athletes not reporting the use of any supplements (Wardenaar et al., 2021).

Was the Myth busted?

- No, the myth is equivocal, although plausible, given more research is needed.
- Explanation: Accurate hydration assessment is important as body fluid balance is essential to optimize health and performance. It is suggested vitamin supplementation might change Uc but not the interpretation of hydration status practically. Previous studies suggested vitamin supplementation did not change the accuracy of hydration assessment while using Uc. More research investigating the effect of vitamin intake on urine color might be required in dehydrated subjects.
- Suggested reading: Kenefick et al. (2015).

What Is the Impact of Preexercise Food-Sourced Calcium Intake on Acute Bone Metabolism in Response to Exercise?

Louise Burke, PhD, gave an answer to this statement by addressing the following proposition: "You can't get enough calcium from dairy if you want to target pre-exercise Ca supplementation."

Although many athletes have poor bone health, some risk factors such as nonweight bearing exercise or low energy availability associated with optimal racing physique are endemic to their sport. Therefore, a preexercise calcium (Ca) supplement might provide a low-cost and practical strategy to contribute to bone health in other ways. Indeed, consumption of a 1,000 mg Ca supplement just before endurance exercise can attenuate the (as yet unexplained) drop in ionic Ca at the onset of exercise and potentially reduce the subsequent parathyroid response, which releases bone Ca to restore blood Ca homeostasis (Barry et al., 2011). Several studies in high-performance athletes have investigated whether a suitable gut release of Ca could be achieved through dietary (dairy-based) means and thus spare bone breakdown. The first compared a dairy-based breakfast (~1,300 mg Ca, three serves of dairy foods) and an energy-matched low Ca breakfast, consumed 90 min prior to a lab-based endurance cycling protocol in Tier 3–4 female cyclists (Haakonssen et al., 2015). The dairy-rich meal helped to stabilize serum ionic Ca, and reduced the increase in parathyroid hormone and C-terminal telopeptide of type I collagen (CTX-I; marker of bone resorption/breakdown) observed in the control trial in the postexercise period (Haakonssen et al., 2015). Importantly, the dairy-rich meal did not increase self-reports of gastrointestinal discomfort by cyclists or affect performance of a time-trial within the workout (Haakonssen et al., 2014). A similar finding was reported when Tier 3–4 male rowers undertook two nonweight bearing (row or bike ergometer) workouts in succession, with 2.5 hr between sessions, and each preceded by a high dairy (~1,000 mg Ca) or isocaloric low Ca meal (Lundy et al., 2023). Again, the consumption of food Ca sources, 2 hr prior to the training session, created a cumulative and prolonged stabilization of blood ionic Ca during exercise and, in turn, reduced the postexercise parathyroid response and partially attenuated the increase in the bone resorption biomarker (Lundy et al., 2023).

Was the Myth busted?

- Yes, the myth, as stated, is busted.
- Explanation: Some unexplained mechanism seems to cause a fall in blood Ca concentrations during exercise, requiring bone to sacrifice some of its Ca content to prop up these important levels. Consuming a source of Ca—as a supplement or in food form—appears to reduce this effect by providing a gut source of Ca to spare bone breakdown. While this looks promising as an acute situation, we are yet to see if repeated practice leads to improved bone health.
- Suggested reading: Lundy et al. (2023).

Conclusions

The responses provided are intended to stimulate discussion among sports nutrition practitioners and researchers. They reflect individual perspectives on specific propositions and do not represent a comprehensive overview of the existing literature. As such, each answer should be interpreted within the context of the question it was designed to address. The quick takeaways of this article are organized within four combined categories, that is, energy and fueling, micronutrients, hydration, and nutrients promoting muscle growth and/or maintenance. Starting with energy and fueling:

- In the context of athletes, NSSs are not well studied. They may provide some benefit in specific situations, for example, for athletes with tight calorie budgets who struggle to reduce free sugar intake. Pending data confirming efficacy in athletic populations, however, they are not recommended as a primary energy or weight management strategy.
- During, but also after exercise, commercially available CHO drinks and foods can be replaced by a food-based approach supporting fueling and glycogen repletion.

The nutrient content of foods is important to consider, and this session focused on what defines if a food is rich in a certain micronutrient and if foods can provide clinically effective doses of micronutrients to replace supplementation:

- Foods can be considered rich in Fe when containing ≥20% of the recommended daily value from a standard serving. The Fe content of teff is often high due to soil contamination, and based on the level of contamination, the grain can or cannot be considered Fe-rich.
- It is possible, although not always practical, to consume enough dairy before endurance exercise to support markers

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of bone metabolism. Whether this contributes to long-term bone health is yet to be determined.

Optimizing hydration is important for general health and everyday foods can contribute to maintaining a sufficient fluid balance:

- Commercially available sports drinks are convenient, but depending on individual needs homemade drinks may theoretically have a similar impact on meeting hydration and fueling goals.
- For those looking for a more "natural" rehydration beverage, coconut water can be a good option for rehydration due to its flavor. Sodium should be added to coconut water to mimic the content of regular electrolyte drinks.
- To ensure optimal hydration, it is often recommended to monitor hydration, for example, by assessing Uc as urine concentration and color are positively associated with each other. Vitamin intake, in particular B vitamins, could influence Uc, as urine samples with vitamin B2 and B12 can appear to have a darker color. However, more research is needed in the dehydrated state.

Finally, the session also addressed whether nutrients promoting muscle growth and/or maintenance coming from sports foods could be replaced by everyday foods:

- Vegetarian diets contain little to no creatine, and recent data show that a low-meat omnivorous diet also provides inadequate amounts of creatine. Supplementation can improve sports performance and health.
- The intake of everyday foods (such as meat and dairy) can provide sufficient amounts of BCAAs, or leucine in particular, but the total energy intake of the food to meet the leucine threshold must be considered. While supplementation is a feasible option, a complete protein such as whey protein is preferable, specifically when prioritizing lower energy intake.
- Vegetarians and vegans, but also omnivores, do not need to supplement beta-alanine or carnitine; however, targeted supplementation could lead to performance benefits as shown in athletes on an omnivorous regular diet.

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Nontechnical Summary

This article summarizes the 2025 American College of Sports Medicine "10 Questions/10 Experts" session hosted by Professionals in Nutrition for Exercise and Sports, where 10 nutrition researchers and/or practitioners addressed current myths in sports performance through rapid-fire verdicts: busted, plausible/equivocal, or confirmed. Topics included dietary creatine, nonsugar sweeteners, leucine sources, food-based carbohydrate fueling, homemade hydration solutions, vegetarian micronutrients, teff and iron content, coconut water hydration, urine color accuracy, and preexercise calcium. The dialogue combined emerging and established perspectives, emphasizing practical applications and suggesting that everyday foods often match—or inform—the performance benefits promoted by supplements and sports products.

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